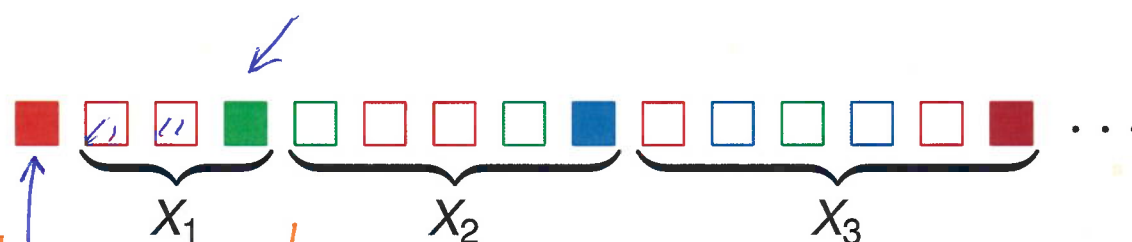


Coupon Collector Problem

There are N types of coupons. Each time one obtains a coupon, it is equally likely to be any one of the N types. Find the expected number of coupons one needs to amass before obtaining a complete set of at least one of each type.

Solution: Let X denote the number of coupons collected before a complete set is attained.



In general, X_i = the number of additional steps needed to see the $(i + 1)$ -st type after having collected distinct i types.

$$\mathbb{E}(X) = 1 + \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_{N-1})$$

$$\frac{N}{N-1}$$

$$\frac{N}{N-2}$$

$$\frac{N}{N-(N-1)}$$

$$X_1 \sim \text{Geometric}\left(\frac{N-1}{N}\right) \quad E(X_1) = \frac{N}{N-1}$$

Observe that $X_i \sim \text{Geometric}\left(\frac{N-i}{N}\right)$ and thus

$$E(X_i) = \frac{N}{N-i}.$$

Therefore

$$\begin{aligned} \boxed{E(X)} &= 1 + E(X_1) + E(X_2) + \cdots + E(X_{N-1}) \\ &= \underline{1} + \frac{N}{N-1} + \frac{N}{N-2} + \cdots + \frac{N}{1} \\ &= N \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} \right) \\ &\approx N \ln N. \end{aligned}$$

Recall the Euler's constant:

$$\lim_{N \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} - \ln N \right).$$

Problem: Expected Number of Runs

- Consider a random sequence of n 1s and m 0s (there are $\binom{n+m}{n} = \frac{(n+m)!}{n!m!}$ possible such sequences and we pick one uniformly).
- Any consecutive string of 1s is said to constitute a run of 1s. (For example, 1110110010 has 3 runs of 1s)
- Let R be the number of runs of 1s. 111 --- 1 0 --- 0 0
- Find $\mathbb{E}(R)$.

Let I_i be the indicator variable that a run of 1s starts at the i -th position.

Then index

001011
↑ $I_2 = 0$
 I_3

$$\mathbb{E}(R) = \sum_{i=1}^{m+n} \mathbb{E}(I_i)$$

111 --- 1 0 --- 0 11 --- 10 ---
↑ I_1 I_{27}
↑ I_2 takes value 0
↑ I_{15} ↑ I_{27}

$$R = I_1 + I_2 + \dots + I_{m+n}$$

$$\mathbb{E}(R) = \mathbb{E}(I_1 + I_2 + \dots + I_{m+n})$$

$$= \mathbb{E}(I_1) + \mathbb{E}(I_2) + \dots + \mathbb{E}(I_{m+n})$$

$$\mathbb{E}(I_1) = P\{\text{Event 1 happens}\} = P\{\text{a run of 1s starts at the first position}\}$$

$$= \frac{n}{m+n}$$

$$\begin{aligned}
 E(I_2) &= P\{\text{Event 2 happens}\} \\
 &= P\{\text{a run of 1s starts at 2-nd position}\} \\
 &= \frac{m}{m+n} \cdot \frac{n}{(m-1)+n}
 \end{aligned}$$

$$E(I_1) = P\{\text{a run starts at 1st position}\} = \frac{n}{n+m}$$

$$\begin{aligned}
 E(I_i) &= P\{\text{a run starts at } i\text{-th position}\} \\
 &= P\{0 \text{ at } (i-1)\text{-st position and } 1 \text{ at } i\text{-th position}\} \\
 &= \frac{m}{n+m} \cdot \frac{n}{n+m-1}
 \end{aligned}$$

Hence

$$\begin{aligned}
 E(R) &= E(I_1) + E(I_2) + \cdots + E(I_{m+n}) \\
 &= \frac{n}{m+n} + (m+n-1) \frac{mn}{(n+m)(n+m-1)} \\
 &= \frac{n(m+1)}{n+m}
 \end{aligned}$$

Expectation of Products of Independent Random Variables

Proposition

If X and Y are independent, then

$$\mathbb{E}(XY) = \mathbb{E}(X) \mathbb{E}(Y),$$

provided that $\mathbb{E}(X), \mathbb{E}(Y)$ exist.

Handwritten derivation:

$$\begin{aligned} \mathbb{E}(XY) &= \iint_{\mathbb{R}^2} (xy) f(x,y) dx dy \\ &= \iint_{\mathbb{R}^2} (x P_X(x)) (y P_Y(y)) dx dy \\ &= \int_{\mathbb{R}} x P_X(x) dx \int_{\mathbb{R}} y P_Y(y) dy \\ &= \mathbb{E}(X) \cdot \mathbb{E}(Y) \end{aligned}$$

Corollary

If X and Y are independent random variables and g and h are fixed functions, then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \cdot \mathbb{E}[h(Y)]$$

provided that the expectations on the right-hand side exist.

MH2500 Probability and Introduction to Statistics

Slides for Week 10-2

Topics: Covariance

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Covariance – Definition

- The variance of a random variable measures its variability.
- The **covariance** of two random variables measures their joint variability, or **degree of association**.

Definition

If X and Y are jointly distributed random variables with expectations μ_X and μ_Y , respectively, then the covariance of X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)],$$

provided that the expectation exists.

$[Y=X]$ $\text{Cov}(X, X) = \mathbb{E}[(X - \mu_X)(X - \mu_X)]$
 $= \mathbb{E}(X^2 - 2\mu_X X + \mu_X^2)$
 $= \mathbb{E}(X^2) - (\mu_X)^2$

Covariance is the average value of the product of the deviation of X from its mean and the deviation of Y from its mean.

$\text{Var}(X)$

- Positive covariance means both X and Y tend to be larger than their respective means or both X and Y tend to be smaller than their respective means. ✓
- Negative covariance means X tends to be larger (smaller) than its mean while Y tends to be smaller (respectively larger) than its mean. ✓

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Proof:

$$\parallel$$

$$\text{Cov}(Y, X)$$

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mu_Y - \mathbb{E}(Y)\mu_X + \mu_X\mu_Y \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y). \quad \checkmark \end{aligned}$$

So $\text{Cov}(X, X) = \text{Var}(X)$. $(XY)(\omega) = X(\omega)Y(\omega) = Y(\omega) \cdot X(\omega)$

$$XY = YX$$

$$E(XY) = E(X)E(Y)$$

- If X and Y are independent, then $E(XY) = E(X)E(Y)$ and $Cov(X, Y) = 0$.

- The converse is not true. //

$$Cov(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent.}$$

Example

Let X be uniform on $\{-1, 0, 1\}$ and $Y = \mathbf{1}_{\{X=0\}}$.

- It is always true that $XY = 0$, and thus $E(XY) = 0$.
- $E(X) = 0$ hence $Cov(X, Y) = E(XY) - E(X)E(Y) = 0$ but X and Y are not independent.

$$XY(\omega) = X(\omega)Y(\omega)$$

$$\begin{array}{ll} X(\omega) = \begin{cases} 1 \\ -1 \end{cases} & Y(\omega) = 0 \\ X(\omega) = 0, & Y(\omega) = 1 \end{array}$$

$$XY = 0$$

$$E(X) = \frac{1}{3}(-1) + \frac{1}{3} \times 0 + \frac{1}{3}(1) = 0.$$

Covariance of linear combinations

- Recall that $\mathbb{E}(a) = a$. Hence

$$\begin{aligned}\text{Cov}(a, Y) &= \mathbb{E}[(a - \mathbb{E} a)(Y - \mathbb{E} Y)] \\ &= \mathbb{E}[\underline{0} \cdot (Y - \mathbb{E} Y)] \\ &= 0.\end{aligned}$$

- Recall $\mathbb{E}(aX) = a\mathbb{E}(X)$. Hence

$$\begin{aligned}\text{Cov}(aX, Y) &= \mathbb{E}[(aX - \mathbb{E}(aX))(Y - \mathbb{E} Y)] \\ &= \mathbb{E}[(aX - a\mathbb{E} X)(Y - \mathbb{E} Y)] \\ &= a\mathbb{E}[(X - \mathbb{E} X)(Y - \mathbb{E} Y)] \\ &= a\text{Cov}(X, Y).\end{aligned}$$

$$\begin{aligned}
Cov(X + Y, Z) &= \mathbb{E}[(X + Y - \mathbb{E}(X + Y))(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X + Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X)(Z - \mathbb{E} Z) + (Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X)(Z - \mathbb{E} Z)] + \mathbb{E}[(Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= Cov(X, Z) + Cov(Y, Z).
\end{aligned}$$

Therefore $Cov(\cdot, \cdot)$ is linear w.r.t. the first argument:

$$Cov(aX + bY, Z) = aCov(X, Z) + bCov(Y, Z).$$

and same w.r.t. the second argument since $Cov(\cdot, \cdot)$ is symmetric.

Combine the above, $Cov(aW + bX, cY + dZ)$

$$\begin{aligned}
&= Cov(aW, cY) + Cov(bX, cY) + Cov(aW, dZ) + Cov(bX, dZ) \\
&= acCov(W, Y) + bcCov(X, Y) + adCov(W, Z) + bdCov(X, Z).
\end{aligned}$$

$$\begin{aligned}
& \text{Cov}(U, V) \\
&= \text{Cov}(a + \sum \square, V) \\
&= \text{Cov}(\underline{a}, V) + \text{Cov}(\sum \square, V) \\
&= \underline{0} + \text{Cov}(\sum \square, V)
\end{aligned}$$

Theorem

Suppose that $U = a + \sum_{i=1}^n b_i X_i$ and $V = c + \sum_{j=1}^m d_j Y_j$. Then

$$\text{Cov}(U, V) = \sum_{i=1}^n \sum_{j=1}^m b_i d_j \text{Cov}(X_i, Y_j).$$

Since $\text{Var}(X) = \text{Cov}(X, X)$,

$$\begin{aligned}
\underline{\text{Var}(X + Y)} &= \text{Cov}(X + Y, X + Y) \\
&= \underline{\text{Cov}(X, X)} + \underline{\text{Cov}(X, Y)} + \underline{\text{Cov}(Y, X)} + \underline{\text{Cov}(Y, Y)} \\
&= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).
\end{aligned}$$

when X, Y independent

Corollary 1

$$\text{Var} \left(a + \sum_{i=1}^n b_i X_i \right) = \sum_{i=1}^n \sum_{j=1}^n b_i b_j \text{Cov}(X_i, X_j).$$

Corollary 2

Suppose the X_i 's are independent. Then

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i).$$

Example: Sample Variance

Let $X_1, \dots, X_n$ be i.i.d. random variables having expected value μ and variance σ^2 , and let $\bar{X} = (\sum_i X_i)/n$ be the sample mean.

The **sample variance** is defined as

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$\underline{\underline{\mathbb{E}(S^2)}} = \frac{1}{n-1} \sum_{i=1}^n \mathbb{E}[(X_i - \bar{X})^2] = \underline{\underline{\sigma^2}}.$$

Proof:

$$\begin{aligned} \underline{\underline{\mathbb{E}[(X_i - \bar{X})^2]}} &= \mathbb{E}((X_i - \underline{\underline{\mu}} + \underline{\underline{\mu}} - \bar{X}))^2 \\ &= \mathbb{E}\left((X_i - \mu)^2 + 2(X_i - \mu)(\mu - \bar{X}) + (\mu - \bar{X})^2\right) \\ &= \mathbb{E}((X_i - \mu)^2) + 2\mathbb{E}((X_i - \mu)(\mu - \bar{X})) + \mathbb{E}((\mu - \bar{X})^2) \\ &= \underline{\underline{\sigma^2}} - 2\underline{\underline{\text{Cov}(X_i, \bar{X})}} + \underline{\underline{\text{Var}(\bar{X})}} \\ &= \sigma^2 - 2\text{Cov}\left(X_i, \frac{1}{n} \sum_j X_j\right) + \text{Var}\left(\frac{1}{n} \sum_j X_j\right) \\ &= \sigma^2 - \frac{2}{n} \sum_j \underline{\underline{\text{Cov}(X_i, X_j)}} + \frac{1}{n^2} \sum_j \underline{\underline{\text{Var}(X_j)}} \quad \swarrow X_1, \dots, X_n \text{ are independent} \\ &= \sigma^2 - \frac{2}{n} \underline{\underline{\text{Var}(X_i)}} + \frac{1}{n^2} \cdot \underline{\underline{n\sigma^2}} = \underline{\underline{\frac{n-1}{n}\sigma^2}}. \end{aligned}$$

$$\begin{aligned} i \neq j &\Rightarrow X_i, X_j \text{ independent} \\ &\Rightarrow \text{Cov}(X_i, X_j) = 0 \\ E(S^2) &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{n-1}{n} \sigma^2\right) \\ &= \frac{1}{n-1} \cdot (n-1) \sigma^2 \\ &= \sigma^2. \end{aligned}$$

Variance of $\text{Bin}(n, p)$

Example

Suppose $X_1, X_2, \dots, X_n$ are independent Bernoulli random variables with $P\{X_i = 1\} = p$. Let $Y = X_1 + X_2 + \dots + X_n$. Find $\text{Var}(Y)$.

For a Bernoulli random variable,

$$\text{Var}(X_i) = p(1 - p).$$

$$E(X_i) = 1 \cdot P\{X_i = 1\} + 0 \cdot P\{X_i = 0\} = p.$$

Hence

$$\text{Var}(Y) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

Correlation coefficient

Definition

If X and Y are jointly distributed random variables and the variance and covariance of both X and Y exist and the variances are nonzero, then the correlation of X and Y , denoted by ρ , is

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

- Both numerator and denominator have the same units and so correlation has no units.
- Suppose both X and Y are subject to linear transformations, both Variance and Covariance changes but ρ does not change.
- Since ρ does not depend on the units of measurement, it is usually a more useful measure of association.

Denote by $\sigma_{XY} = \text{Cov}(X, Y)$. Then

$$\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y},$$

i.e.,

$$\sigma_{XY} = \rho \sigma_X \sigma_Y.$$

Cauchy-Schwarz Inequality

We always have $-1 \leq \rho \leq 1$. Furthermore, $\rho = \pm 1$ if and only if $P\{Y = a + bX\} = 1$ for some constants a and b .

Proof:

$$\begin{aligned} 0 &\leq \text{Var} \left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y} \right) \\ &= \text{Var} \left(\frac{X}{\sigma_X} \right) + \text{Var} \left(\frac{Y}{\sigma_Y} \right) + 2\text{Cov} \left(\frac{X}{\sigma_X}, \frac{Y}{\sigma_Y} \right) \\ &= \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} + 2 \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= 1 + 1 + 2\rho. \end{aligned}$$

Hence $\rho \geq -1$.

$$\boxed{-1 \leq \rho \leq 1}$$

A similar argument shows that

$$0 \leq \text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right) = 2 - 2\rho,$$

which shows that $\rho \leq 1$.

Suppose $\rho = 1$. Then

$$\text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right) = 0,$$

which implies that

$$P \left\{ \frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = c \right\} = 1$$

$$P \left\{ \omega: \frac{X(\omega)}{\sigma_X} - \frac{Y(\omega)}{\sigma_Y} = c \right\} = 1$$

for some constant c . This is equivalent to $P\{Y = a + bX\} = 1$ for some constants a and b .

$$Y = \frac{\sigma_Y}{\sigma_X} X - \frac{c}{\sigma_X}$$

- Similar argument for $\rho = -1$.

Conditional Expectation and Prediction

We discuss the conditional mean associated to a conditional distribution.

Definition

Suppose X and Y are **discrete** random variables and that the conditional frequency function of Y given x is $p_{Y|X}(y|x)$. The **conditional expectation** of Y given $X = x$ is

$$\mathbb{E}(Y|X = x) = \sum_y y p_{Y|X}(y|x).$$

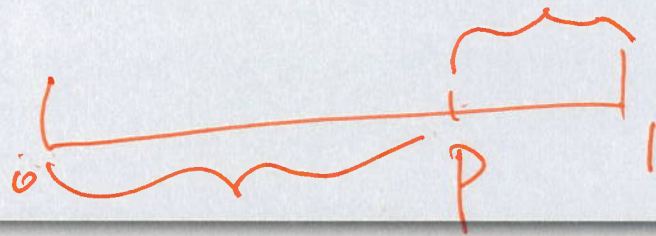
For the **continuous** case, we have

$$\mathbb{E}(Y|X = x) = \int y f_{Y|X}(y|x) dy.$$

Example

Consider a Poisson process on $[0, \infty)$ with mean λ , and let N be the number of events in $[0, 1]$. For $p \in (0, 1)$, let X be the number of events in $[0, p]$.

- Find $\mathbb{E}(X|N = n)$.



- From the assumption of a Poisson process, the count in the two intervals are independent Poisson random variables with parameters $p\lambda$ and $(1-p)\lambda$.
- We proved before that the conditional distribution of X given $N = n$ is Bin(n, p). Hence $\mathbb{E}(X|N = n) = np$.
- We also write $\mathbb{E}(X|N) = Np$. Hence $\mathbb{E}(X|N)$ is a function of N .
- In a similar way, for any two random variables X and Y , $\mathbb{E}(Y|X)$ is a random variable as a function of X .

$$\underline{E(X|N=n) = np.}$$

Example

The joint density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-x/y} e^{-y} / y, & 0 < x < \infty, 0 < y < \infty; \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{E}(X|Y = y)$ (assuming that $y > 0$).

We calculated before that

$$f_{X|Y}(x|y) = \frac{1}{y} e^{-x/y},$$

that is, the conditional distribution of X given $Y = y$ is $\text{Exp}(1/y)$.

Hence $\mathbb{E}(X|Y = y) = y$.

Conditional Expectation

Just as conditional probabilities satisfy all of the properties of ordinary probabilities, so do conditional expectations satisfy the properties of ordinary expectations. For instance,

$$\mathbb{E}(\underline{g(Y)} | \underline{X = x}) = \begin{cases} \sum_y \underline{g(y)p_{Y|X}(y|x)} & \text{in the discrete case} \\ \int_{-\infty}^{\infty} \underline{g(y)p_{Y|X}(y|x)dy} & \text{in the continuous case} \end{cases}$$

and

$$\mathbb{E} \left(\sum_{i=1}^n Y_i \middle| X = x \right) = \sum_{i=1}^n \mathbb{E}(Y_i | X = x)$$

- Conditional expectation given $X = x$ can be thought of as being an ordinary expectation on a **reduced sample space** consisting only of outcomes for which $X = x$.

Law of total expectation

The mean of $\mathbb{E}(Y|X)$ is given by $\mathbb{E}[\mathbb{E}(Y|X)]$.

Theorem

$$\mathbb{E}(Y) = \mathbb{E}_X[\mathbb{E}_Y(Y|X)].$$

Handwritten notes: "function" with an arrow pointing to $\mathbb{E}_Y(Y|X)$, and a bracket under $\mathbb{E}_X$.

$$\mathbb{E}(Y) = \begin{cases} \sum_x \mathbb{E}(Y|X=x)p_X(x) & \text{in the discrete case} \\ \int_{-\infty}^{\infty} \mathbb{E}(Y|X=x)p_X(x)dx & \text{in the continuous case} \end{cases}$$

To calculate $\mathbb{E} Y$, we may take a weighted average of the conditional expected value of Y given that $X = x$, each of the terms $\mathbb{E}(Y|X = x)$ being weighted by the probability of the event on which it is conditioned.

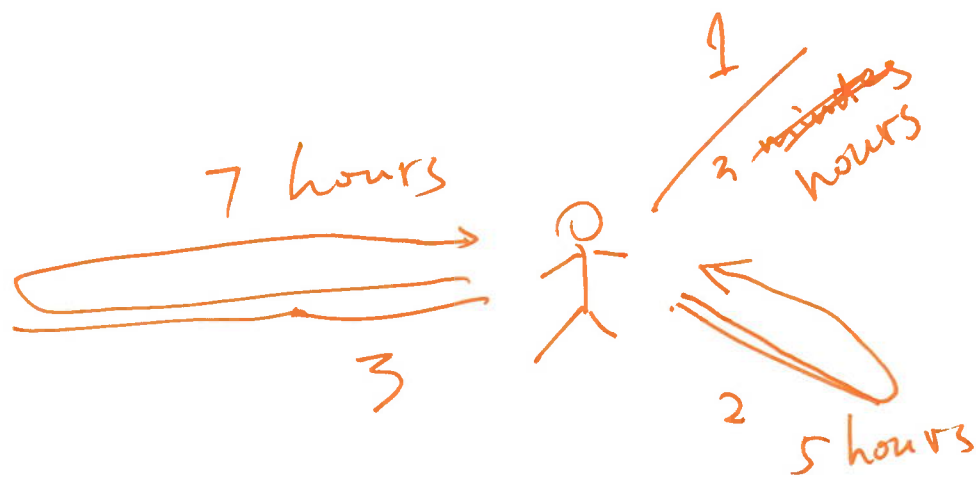
Proof: We only prove the discrete case. The continuous case is similar.

$$\begin{aligned} \mathbb{E}[\mathbb{E}(Y|X)] &= \sum_x \mathbb{E}(Y|X=x) p_x(x) \\ &= \sum_x \left(\sum_y y p_{Y|X}(y|x) \right) p_x(x). \end{aligned}$$

Changing the order of the summation gives

$$\mathbb{E}[\mathbb{E}(Y|X)] = \sum_y y \sum_x p_{Y|X}(y|x) p_x(x) = \sum_y y p_Y(y),$$

by the law of total probability. Therefore, $\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}(Y)$.



Example

A miner is trapped in a mine containing 3 doors, each has a tunnel behind.

The $\begin{Bmatrix} \text{first} \\ \text{second} \\ \text{third} \end{Bmatrix}$ door $\begin{Bmatrix} \text{leads to safety} \\ \text{returns to the mine} \\ \text{returns to the mine} \end{Bmatrix}$ after $\begin{Bmatrix} 3 \\ 5 \\ 7 \end{Bmatrix}$ hours of travel.

If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

Solution: Let X be the expected escape time. Then

$$\mathbb{E}(X|\text{first}) = 3, \quad \mathbb{E}(X|\text{second}) = 5 + \mathbb{E}(X), \quad \mathbb{E}(X|\text{third}) = 7 + \mathbb{E}(X)$$

$$E(Y) = E_X(\underline{E_Y(Y|X)})$$

On the other hand,

$$\underline{E(X)} = \underline{E(X|\text{first})} \cdot \frac{1}{3} + E(X|\text{second}) \cdot \frac{1}{3} + E(X|\text{third}) \cdot \left(\frac{1}{3}\right)$$

$$= 5 + \frac{2}{3} E(X)$$

$$\Rightarrow \underline{E(X) = 15}$$

||
 $3 * \frac{1}{3} + (5 + E(X)) \frac{1}{3}$
 $+ (7 + E(X)) \frac{1}{3}$
 $= \boxed{1 + \frac{5}{3} + \frac{7}{3}} + \frac{2}{3} E(X)$
 $\quad \quad \quad 5$

Example

Suppose the random variable N denotes the number of jobs in a single-server queue and X_i the service time for the i -th job. Let T denote the time to serve all jobs in the queue, i.e.,

$$T = X_1 + X_2 + \cdots + X_N.$$

Prove that $\mathbb{E}(T) = \mathbb{E}(N) \mathbb{E}(X)$. Does this agree with intuition?

By the Theorem, $\mathbb{E}(T) = \mathbb{E}[\mathbb{E}(T|N)]$.

Since $\mathbb{E}(T|N = n) = n \mathbb{E}(X)$, we see that $\mathbb{E}(T|N) = N \mathbb{E}(X)$ and so

$$\mathbb{E}(T) = \mathbb{E}[N \mathbb{E}(X)] = \mathbb{E}(N) \mathbb{E}(X).$$

This agrees with the intuitive guess that the average time to complete N jobs where N is random, is the average value of N times the average amount of time to complete a job.

